Revisit: Did It Change, or Did We Just Not Look?

Amitabh Mahapatra

***Abstract*—Two casual phone scans of a room disagree for many reasons besides change: the second scan looks from elsewhere, skips a corner, or finds the twin of a chair that moved. We present Revisit, a weekend-built pipeline that reconstructs both scans jointly with a feed-forward 3D model, proposes object-level changes, and reports a proposal only after two checks. An evidence ledger marks objects the second scan never observed as UNVERIFIED and suggests where to stand to settle them with a re-look. A vision-language verifier, trained with multi-reward reinforcement learning that includes a calibration reward, judges the rest and returns a probability of change. On the SceneDiff test split the trained verifier raises pixel IoU from 34.5 to 40.5 over its zero-shot base and the expected calibration error of its fused probability falls from 0.377 to 0.066. In a room with nine staged changes, the system with a re-look gets four to five changes fully right and three partly right, but it also reports six to seven changes that never happened, and it accepts 10 to 11 of 17 proposals on two scans of the unchanged room. We describe the system, its training on six 8-GPU machines per verifier, and what a reliable version would take.**

I. Introduction

A robot that returns to a room, or a person checking on one, wants a short list: what moved, what appeared, what is gone. Phone scans are the cheapest source of 3D evidence for that list, and LiDAR phones add metric depth and a camera path for free. Finding differences between two scans is easy. Deciding which ones to believe is the problem: the second scan sees the room from other places, misses corners and meets look-alike objects, and each of these creates differences that are not changes (Fig. 1).

Revisit treats change detection as propose, then verify (Fig. 2). It reconstructs the keyframes of both scans in one frame with Pi3 [2], sets metric scale from the ARKit camera path, and proposes changed objects with the SceneDiff method [1]. Each proposal then faces two checks. The first concerns observability: if the second scan never pointed a camera at an object's old place, the object is UNVERIFIED, not removed, and the system proposes a pose to look from; a re-look scan joins the reconstruction and settles it. The second concerns appearance: an 8B vision-language model compares crops from both scans, reads the 3D scores and answers moved, added, removed or none with a confidence. We train this verifier with GDPO [3] and the calibration reward of RLCR [4], so that its probability can drive decisions such as when to ask for another look.

Our contributions are:

- a propose-then-verify pipeline for two phone scans with an evidence ledger that separates changed, unchanged and unobserved, plus tests for replaced and swapped objects;
- a verifier trained with multi-reward RL at two scales, 45 steps on one 8-GPU machine and about 15 hours on six 8-GPU machines, whose fused probability reaches an expected calibration error (ECE) of 0.066 on SceneDiff, against 0.377 zero-shot;
- a small adversarial room benchmark with twins, a swap, a replacement, a flat object and a hidden corner, audited change by change, with an analysis of where the system fails.

Code, adapters and captures: https://github.com/amitabhm1/revisit.

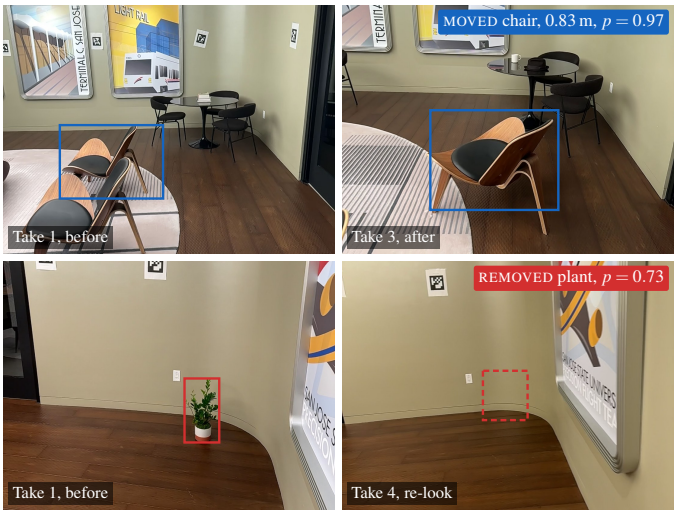


Fig. 1. Changed, or never seen? Top: one of two identical rug chairs was pulled back between the scans, and Revisit reports a move of 0.83 m with $p_{\text{chg}} = 0.97$. Bottom: the after scan never filmed this corner, so it holds no evidence about the plant; the re-look does, and the plant is reported removed ($p_{\text{chg}} = 0.73$). Boxes come from the system output (Qwen verifier).

II. Related Work

Change detection. Image-pair methods localize differences in 2D [5] and degrade under large viewpoint changes. Robotic mapping tracks object-level change in volumetric maps built from posed RGB-D streams [6], [7], and ChangeSim [8] benchmarks online change detection for robots in simulation. MV3DCD [9] localizes change across unposed views without labels. SceneDiff [1] studies object change between two videos taken along different trajectories, releases a benchmark of 350 video pairs and a training-free method built on Pi3, SAM and DINOv3; we use its method for proposals and its benchmark for training and evaluation.

Reconstruction and segmentation. DUSt3R [10] and VGGT [11] predict geometry from unposed images; Pi3 [2] drops the reference view through permutation equivariance, which suits frames from two independent scans. SAM [12] segments from point prompts, SAM 2 [13] extends it to video with a memory, and SAM 3 [14] adds concept prompts with an instance detector and tracker.

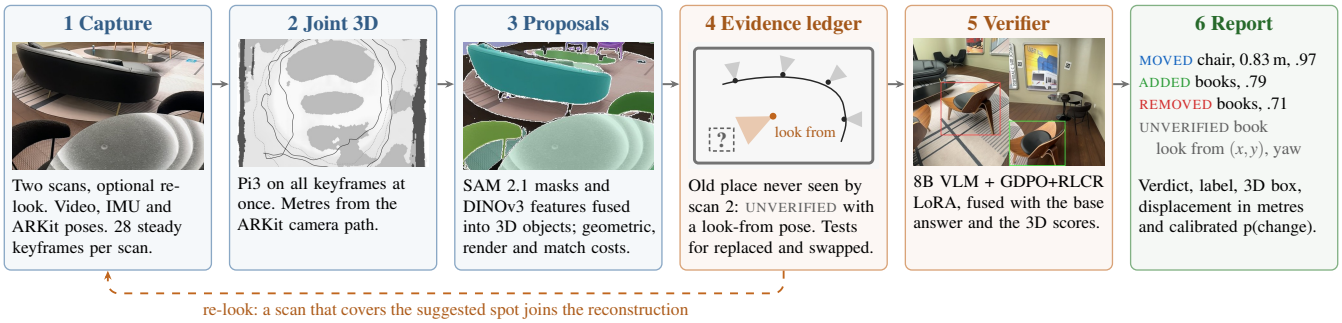


Fig. 2. Revisit in six stages. Stages 1 to 3 build one metric reconstruction of both scans and propose changed objects. Stages 4 and 5 decide which proposals to believe: the ledger handles what the second scan never saw, the verifier handles what it did see. Report lines are from the before/after run with the Qwen verifier. A comparison takes 9 to 13 minutes on one H100.

VLM verifiers trained with RL. Qwen3-VL [15] and SenseNova-SI [16], a variant tuned for spatial reasoning, are our base models. GRPO [17] normalizes rewards within a group of sampled answers; GDPO [3] normalizes each reward separately before summing, so a weak reward is not drowned out; RLCR [4] adds a Brier-score reward that makes stated confidence calibrated. We combine the last two.

III. Method

A. Capture and joint reconstruction

Revisit first ran on plain video, and that path remains: SceneDiff provides only RGB, so all SceneDiff training and evaluation use every 30th frame and work without metric scale. For the room we recorded with Stray Scanner on an iPhone with LiDAR, which stores 256×192 depth with confidence, the ARKit camera path and the IMU next to the 30 fps video. Two of these extras are used: the gyroscope picks keyframes and the ARKit path sets metric scale. From each scan we keep 28 keyframes, one per time window, at the lowest rotation rate in the window, which avoids motion blur without image analysis. Pi3 [2] reconstructs the keyframes of both scans (and of a re-look) in a single forward pass, without metric scale; for keyframe pairs within each scan we take the ratio of ARKit to Pi3 camera displacement, with the median over pairs pooled over scans. Per-scan estimates agree within 2 to 5% (8.22 and 8.06 metres per Pi3 unit for the no-change pair).

B. Change proposals

Proposals follow SceneDiff [1]. A grid of point prompts gives SAM masks per keyframe; whole-object and part masks are merged into non-overlapping regions as in LangSplat [18]. Regions are lifted to 3D with Pi3 depth, fused across keyframes into objects, and described with DINOv3 [19] ViT-H+/16 features. Each object then gets three costs against the other scan: a geometric cost from depth disagreement after projecting into the other scan's views, a render cost from DINOv3 feature disagreement after the same projection, and a match cost from the best feature match among the other view's regions. All three exist only where the other scan looked. Objects with high cost become MOVED, ADDED or REMOVED proposals with a mean and a maximum score. We use SAM 2.1 [13]; Sec. V-C explains why.

C. Evidence ledger

Three rules turn geometry into verdict types. (i) A REMOVED object whose centroid lies inside fewer than 12% of the second scan's camera frustums (or whose visible coverage is below 0.14) becomes UNVERIFIED, with a look-from pose 0.85 m from the object on the side of the nearest second-scan camera. (ii) A MOVED object that stayed within 0.28 m but changed appearance (render cost above 0.22, DINOv3 similarity below 0.82) becomes REPLACED. (iii) Two moved objects whose start and end points trade places within 0.45 m become SWAPPED. With a re-look, its keyframes join the reconstruction and the visibility test is replaced by the evidence of the new scan. The ledger sees proposals only, a limit that Sec. V-B makes concrete.

D. Verifier and fusion

For each remaining proposal the verifier sees an evidence card: a crop from each scan, a side-by-side context image with the object boxed in both, and a text block with the proposal type, the three costs, the 3D displacement and the visibility coverage. It answers in a short JSON with a verdict among moved, added, removed and none, a confidence, a label and a one-line reason. Base models are Qwen3-VL-8B-Instruct [15] and SenseNova-SI-1.3-8B [16], each with a LoRA adapter [20] on all attention and MLP projections. A fixed rule combines the adapter's answer, the base model's answer (same weights, adapter disabled) and the 3D scores into the final verdict and p_{chg}: the 3D score gates which answer is trusted, and a logistic on a weighted sum of scores and answer probabilities gives p_{chg}. Its constants were set per RL variant on SceneDiff validation, before any test run.

IV. Training the Verifier

Data. Cards come from Stage-1 proposals on the 105 SceneDiff validation scenes; the 249 scored test scenes are never used. A proposal is labelled with the verdict of the ground-truth object it overlaps with stacked-mask IoU above 0.5, or none.

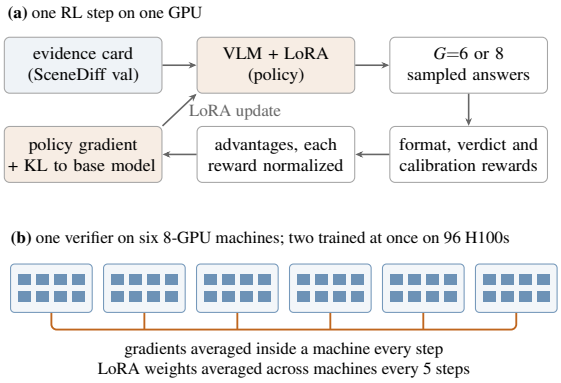


Fig. 3. Verifier training. (a) Each GPU takes one evidence card per step, samples G answers and scores each with three rewards; GDPO normalizes every reward within the group before summing. (b) Each room verifier trained on six 8-GPU machines with periodic weight averaging across machines.

Rewards. The format reward checks for a parseable answer and a short reason that cites geometry, appearance and visibility. The verdict reward gives $+1$ for an exact verdict, $+0.35$ for detecting a change of the wrong type, -0.6 for a missed change, and -1 for a hallucinated change whose 3D cost is low (-0.65 otherwise). The calibration reward is $1-\big(0.6\,\mathrm{Brier}(p_{\mathrm{chg}},\mathrm{changed})+0.4\,\mathrm{Brier}(\mathrm{conf},\mathrm{correct})\big)$. GRPO sums format and verdict (weights 0.25 and 1) and normalizes once; GDPO normalizes each reward within the group first; GDPO+RLCR adds the calibration reward with weight 0.8. The loss is a policy gradient with a per-token KL penalty (β=0.02) to the base model, computed by disabling the adapter.

Two scales. The three SceneDiff adapters each trained on one machine with 8 H100s: one card per GPU per step, G=6, rank 16, learning rate 1.5×10^{-5}, 45 steps over the 938 validation cards in split order (170 of them real changes), in 7.9 minutes. The two room verifiers used a larger pool: the SceneDiff proposals plus four Stage-1 variants on the same validation scenes (Pi3X with DINOv3-H+, DINOv3-L or DINOv2-B and SAM 1, and Pi3X with SAM 3), balanced to 2,748 cards with two hard negatives per real change. Each trained with G=8, rank 32 and a learning rate of 3×10^{-5} decayed by cosine to a quarter over the first 1,500 steps, with photometric and score jitter on the cards, on six 8-GPU machines (Fig. 3b). This is periodic model averaging in the spirit of local SGD [21]. A step took 22 to 24 s. The released adapters are step 2,460 for Qwen and step 2,290 for SenseNova, about 15 hours of training each. Stage 1 for the 355 scenes of both splits, the four extra variants and every verifier evaluation ran sharded over the same 96 GPUs.

V. Experiments

A. SceneDiff

We score the official SceneDiff test metrics on 249 scenes (one of the 250 lost its Stage-1 output in our run) and per-proposal AUROC, 4-way verdict accuracy, Brier score and ECE over 2,231 proposals. Every row filters the same proposals (Pi3, DINOv3 ViT-H+/16, SAM 1, as in SceneDiff).

TABLE I. SceneDiff test split. px/im: pixel IoU per image; obj/im and obj/sc: object AP per image and per scene. RL rows use the 45-step adapters.

Verifier	px/im	obj/im	obj/sc	AUROC	Acc.	Brier	ECE
none (Stage 1 only)	30.0	33.5	17.6	0.810	–	0.253	0.379
Qwen3-VL-8B, zero-shot	34.5	33.4	21.0	0.734	56.0	0.376	0.377
SenseNova-SI, zero-shot	34.8	34.1	22.4	0.772	62.1	0.323	0.321
Qwen + GRPO	38.1	**34.9**	22.5	0.861	64.3	0.210	0.328
Qwen + GDPO	39.6	34.6	**22.6**	**0.862**	69.1	0.152	0.224
Qwen + GDPO+RLCR	**40.5**	34.2	22.5	**0.862**	**71.1**	**0.102**	**0.066**

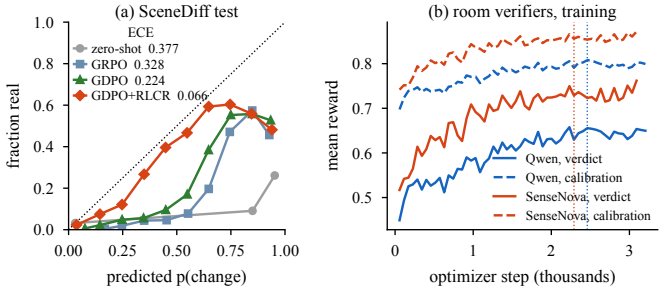


Fig. 4. (a) Reliability on SceneDiff test (bins with at least 15 proposals; ECE in the legend). (b) Mean verdict and calibration rewards of the room verifiers per 57-step window; dotted lines mark the released checkpoints.

Table I shows the pattern. Zero-shot VLMs raise pixel IoU and scene AP but lower AUROC below that of the Stage-1 score: Qwen answers changed with $p_{\mathrm{chg}}\approx0.95$ for 1,083 of the 2,231 proposals, of which 26% are real. Forty-five RL steps fix the ranking (AUROC 0.86 for all three variants) and the variants then differ where their rewards differ. GDPO beats GRPO on accuracy and Brier at the same AUROC, which matches its motivation: summed rewards normalized once let the verdict reward drown the format reward, and separate normalization keeps both. RLCR brings ECE to 0.066 and gives the best pixel IoU. These ECE values are for the fused p_{chg}, set on validation; without a no-adapter control, RL's own share is open. The verifier mostly prunes: GDPO+RLCR keeps 300 of the 327 true proposals and drops 1,168 of the 1,904 false ones, so per-image AP, which depends on proposals Stage 1 never made, stays within 1.5 points. Fig. 4a shows where calibration is earned: below $p_{\mathrm{chg}}=0.65$ the RLCR adapter is calibrated, above it only 48 to 60% of proposals are real.

B. The room benchmark

One furnished room, four captures on an iPhone 14 Pro Max: Take 1 (before, 110 s), Take 2 (before, untouched, different start and direction, 85 s), Take 3 (after, 104 s, with the bottom-left corner kept off screen) and Take 4 (after, 99 s), a re-look that walks the room again and films that corner (Fig. 5). Between Takes 2 and 3 we staged nine changes: (1) a chair moved to another table and turned, (2) a tall stack of books added, (3) one of two identical chairs removed, (4) one of two twin chairs on the rug pulled back, (5) a flat item under 1 cm removed from the rug, (6) a cup moved to another table, (7) a plant removed from the corner, (8) a pillow replaced by a different one, and (9) a hat and a stack of books swapped between tables. Controls that never moved include a decoy hat close

TABLE II. Room benchmark, final configuration (SAM 2.1, 28 keyframes), audited per change. B/A: Take 1 vs. Take 3; +RL: with the Take 4 re-look. False reports: active verdicts that match no staged change.

	Qwen3-VL-8B		SenseNova-SI-8B	
	B/A	+RL	B/A	+RL
Right, of 9	2	4	3	5
Partly right	4	3	4	3
UNVERIFIED	1	–	1	–
Missed	2	2	1	1
False reports	3	6	3	7
Keyword score, of 9	7	6	8	7
No-change pair: false, of 17	11		10	

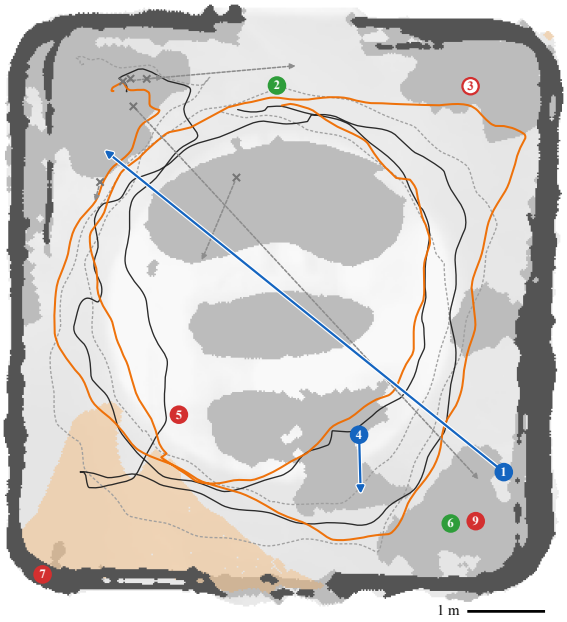


Fig. 5. The test room from above: LiDAR of all takes registered into the joint reconstruction (dark: walls, grey: furniture, light: floor and rug) and the camera paths of Take 1 (dashed), Take 3 (black) and Take 4 (orange); only Take 4 saw the orange area. Circles mark the staged changes at the system's own proposals in the re-look run with Qwen, colored by type: filled when reported, hollow when rejected, with arrows for reported moves. Change 8 has no proposal in this run. Grey marks are the six false reports.

in color to the swapped one, six identical corner chairs, the sofa, the rug and six posters. Distances were not measured, so none are scored. We audited every report against the staged changes: a change is right when an active verdict names the right object with the right type, and partly right when it names the right object with the wrong type, the wrong destination or only half of a swap.

Table II and Fig. 6 summarize the outcome. With the re-look, Qwen gets changes 2, 4, 5 and 7 right, and SenseNova also gets the removed chair (3), which Qwen rejects. Both verifiers get three changes partly right: the cup is reported as added at its new table, the chair of change 1 as moved to a look-alike across the room, and only the books of the swap are reported. The pillow is reported as added before the re-look and has no proposal after it, and no report names the hat.

The ledger works on the flat item: Take 3 saw its place in 1 of 28 keyframes, so it is UNVERIFIED with a look-from pose, and it becomes REMOVED once Take 4 joins ($p_{\text{chg}} = 0.70$ with Qwen). The plant shows the gap in the design. Its corner never appears in Take 3, so the plant gets no costs and no proposal, and the ledger, which sees proposals only, never learns of it; the re-look finds it ($p_{\text{chg}} = 0.73$). The two other UNVERIFIED proposals of the before/after pair are reflections of posters in a glossy table. The keyword score of the repository's scorer over-credits: it counts change 7 for any UNVERIFIED proposal and change 5 for any removed "book", which the swapped books also are.

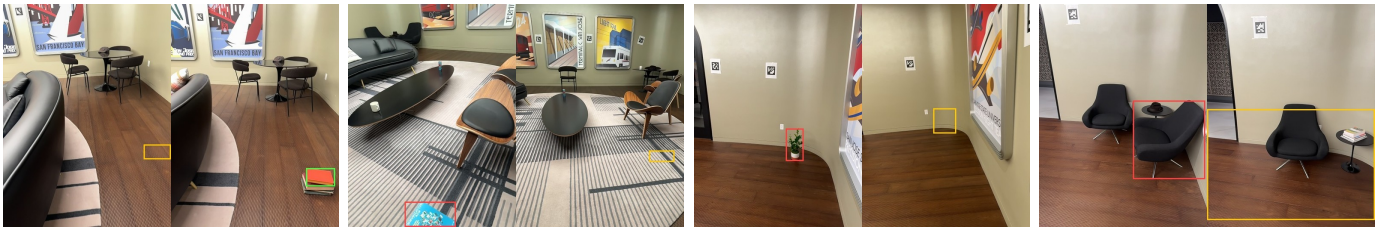
TABLE III. Mask backend and keyframe count. Mask statistics and proposals are for the no-change pair; result columns give Qwen / SenseNova. Found: keyword score (Sec. V-B). Plant: reported removed with the re-look.

Masks	KF	unmasked	3D objects	proposals	false	found	+RL found	plant
SAM 2.1	28	3.6%	14 / 11	17	11 / 10	7 / 8	6 / 7	yes
SAM 1	28	2.4%	15 / 12	20	9 / 10	7 / 7	6 / 6	yes
SAM 2.1	36	3.3%	22 / 18	25	13 / 14	7 / 8	4 / 4	no
SAM 1	36	2.4%	36 / 29	44	25 / 24	7 / 7	6 / 5	no
SAM 3.1	36	24.6%	42 / 37	51	29 / 23	6 / 7	6 / 6	no

The weak spot is the decision that nothing changed. On the no-change pair 10 to 11 of 17 proposals become changes, and with the re-look six to seven false reports sit next to the real ones. They come from the proposal stage in three forms: static objects whose two reconstructions do not line up (the sofa MOVED 1.0 m with $p_{\text{chg}} = 0.84$, the coffee table 0.3 m), fragments that exist in one scan only (a poster and a wall marker, both REMOVED at 0.84), and look-alikes matched across the room. Shown two plausible crops and a 3D story, the verifier agrees, and the calibration it learned on SceneDiff does not carry over. In the room both adapters answer from a template (fixed confidence, label none or box), so labels come from the base model. The 6.6 to 6.8 m moves in Figs. 5 and 6g,h are the clearest case: they all link the top-left and bottom-right corners, whose tables carry identical sets of three chairs, and the no-change pair contains one as well.

C. Masks and keyframes

All three SAM generations were used the same way: as class-agnostic mask generators driven by a grid of point prompts, with the SAM 1 automatic-mask machinery around the point-prompt head of each model. Table III and Fig. 7 show that the number of regions per keyframe is not what separates them. What separates them is consistency across views, because SceneDiff fuses each keyframe's regions into 3D objects and then compares objects between scans. A region that exists in one view and not the next becomes a fragment, and a fragment present in one scan and absent from the other becomes a proposal. On the same keyframes, SAM 3.1 leaves $25 \pm 17\%$ of pixels unmasked (2 to 69% across keyframes) against $3 \pm 2\%$ for SAM 2.1. It builds 42 and 37 objects from a room for which SAM 2.1 builds 22 and 18, and it produces 51 proposals on a pair where nothing changed, twice as many as SAM 2.1. SAM 1 leaves almost no pixel unmasked but still builds 36 and



(a) ADDED stack of books, .79. Change 2.

(b) UNVERIFIED: seen in 1 of 28 after keyframes. With the re-look, REMOVED, .70. Change 5.

(c) REMOVED plant, .73, with the re-look; no proposal without it. Change 7.

(d) Removed chair, change 3: SenseNova reports it (.51), Qwen says none (.34).



(e) ADDED pillow, .67. Change 8 was a replacement: right place, wrong type.

(f) REMOVED books, .71. Half of the swap in change 9; no report names the hat.

(g) MOVED, labelled sign, 6.6 m, .67: a poster that never moved.

(h) MOVED chair, 6.8 m, .91: matched to an identical chair across the room.

Fig. 6. Evidence images from the final Qwen run, before/after unless noted. Each shows the first scan (left, red box) and the second (right, green box, or yellow where the object should be). Top: a correct report, the ledger and the re-look at work, and a change only one verifier accepts. Bottom: the right object with the wrong type or only half of the change, and two errors the verifier accepts with confidence.

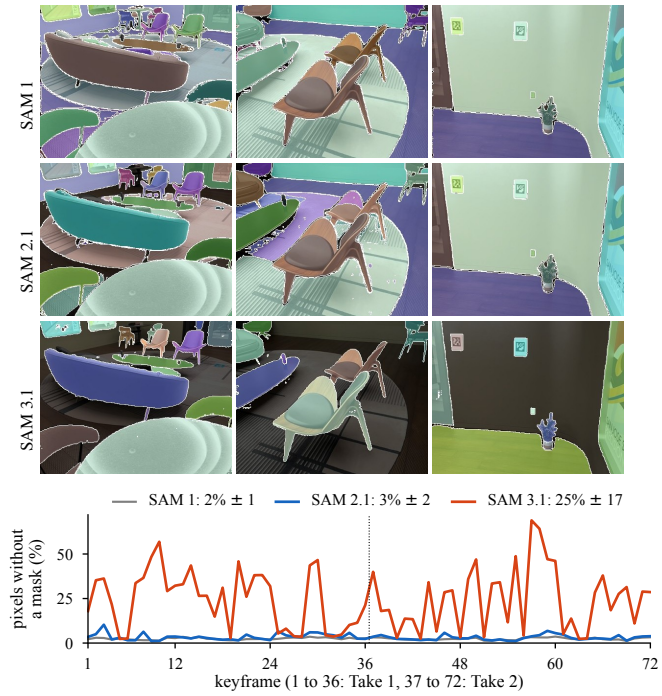


Fig. 7. Masks on the same keyframes of the no-change pair (36 keyframes per scan). Pixels without a mask are darkened. All three backends give 16 to 18 regions per keyframe, but SAM 3.1 drops walls, floor or rug in some views and keeps them in others (bottom: share of unmasked pixels per keyframe, mean ± s.d.).

29 objects; its choice between whole and part masks appears to vary more with the viewpoint.

Why SAM 2.1 fits this use is a hypothesis we have not isolated: its training on video targets masks that stay stable as the camera moves, which multi-view fusion needs, and point prompting is its native mode. SAM 3.1 was queried through its point-prompt head, which bypasses what is new in SAM 3: the concept detector and the identity-preserving tracker. The coverage pattern suggests that its mask-quality scores fall below our filters (predicted IoU 0.70 and stability 0.85 on a 32×32 point grid) on large uniform regions in some views; retuning them on this room would have been tuning on the test set. We expect SAM 3 to win when it is used for what it adds. Prompted with categories or exemplar crops from the first scan, it returns every instance with an identity and tracks it through a video, which targets our two dominant failures: matches to the wrong twin and inconsistency across keyframes.

The keyframe count matters as much as the backend. With 36 keyframes per scan instead of 28, every backend builds more objects from the same room (SAM 2.1: 22 instead of 14 in the first scan), proposes more false changes, and no longer reports the plant after the re-look. More views give more chances to fragment. Both settings were chosen on this room, so Table III is evidence for the mechanism, not a tuned optimum.

Runtime. One comparison with Qwen takes 518 to 682 s of wall-clock time on one H100: keyframes and LiDAR 26 to 36 s; Pi3, masks, features and costs 213 to 271 s (Pi3 itself 14 s, SAM 2.1 masks 72 to 89 s); verification 235 to 335 s, about 14 s per proposal for two unbatched generations.

VI. Limitations and Future Work

Revisit was built in a weekend around one room, so its numbers support a design, not a product. Three parts of that design held up: an explicit ledger that keeps unseen apart from changed, a learned verifier placed after a geometric proposer, and RL with a calibration reward, whose fused probabilities reached an ECE of 0.066 on SceneDiff after 45 steps (0.377 zero-shot). The failures are just as informative, because they

cluster in three places. The system cannot yet decide that nothing changed: it accepts 10 to 11 of 17 proposals on the no-change pair. It does not keep identities: every long false move links two sets of identical chairs. And it does not know what it has not seen: the plant never became a proposal.

The evidence behind these claims is thin. One room, one phone, one person and nine changes cannot establish generalization; keyframe count and mask backend were chosen on the same captures; and our audit judges objects and verdict types, not displacements, which were never measured. The SceneDiff adapters saw 105 validation scenes, fusion constants were set by hand on them, and the room verifiers never trained on a card from a room like ours. The pipeline is also slow: unbatched verification takes half the runtime, and a comparison needs up to 126 GB of host memory.

The goal behind Revisit is larger than comparing two scans. It is a robot that keeps a persistent model of the spaces it works in, updates that model on every visit, attaches a calibrated probability to each belief, and knows which parts are stale and where to look to refresh them. We see four steps toward it, each building on the one before.

Measure the decision that matters. A benchmark of many rooms, repeated no-change visits, several devices and people, and surveyed 3D ground truth for every staged change, scored by false alarms per visit and displacement error rather than recall alone. No-change pairs are cheap to label, since every proposal is false, and they are the hard negatives our verifier lacked. Synthetic edits of real scans and procedural rooms [22] can add changes at scale. This step turns every later idea into a number.

Give objects identities. Replace pairwise matching with a persistent object map: instances segmented and tracked with concept prompts [14], matched across visits as one global assignment with spatial priors, so that a nearby twin always beats a 6.6 m jump. On that map, the proposal stage itself can be learned end to end from synthetic pairs instead of hand-designed costs.

Train the verifier where it runs. Train on-policy, on the cards the full system produces in many rooms, with a set-level reward that fails a whole visit when any change is accepted on a no-change pair, and learn the fusion rule and ledger thresholds we set by hand. The recipe already responds quickly; what it needs is the data from the first step and far more RL compute than we used.

Let the robot decide where to look. Grow the ledger into a coverage map over every object of the previous visit, so that a corner like the plant's is flagged even without a proposal. Plan short looks by expected information gain [23], stop when conformal risk control [24] bounds the false alarms of the reported set, and batch or distill the verifier so the loop runs on the robot.

None of these steps is a small change: each calls for careful design, new data and real compute. Each is also concrete. The three capabilities they rest on, feed-forward multi-view geometry, promptable segmentation that keeps identities, and RL that calibrates a vision-language model, already exist, and this weekend wired off-the-shelf versions of them into one working loop. What remains hard is mostly scale and measurement, where focused effort turns most directly into progress.

VII. Conclusion

Revisit separates three outcomes that naive differencing conflates: changed, unchanged and unseen. A calibrated verifier and an evidence ledger make part of that separation work on SceneDiff and in a staged room. The remaining errors point to identity and coverage as the next problems, and to a persistent, calibrated model of the world as the place where solving them leads.

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